DATA ANALYTICS & BUSINESS INTELLIGENCE PROJECT

DataCo Global Supply Chain

EXCEL | PYTHON | SQL | POWER BI

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1. Executive Summary

This project analyzes the DataCo Supply Chain dataset across 2015–2018 using Python for data preparation and EDA, MySQL for structured business analysis, and Power BI for interactive dashboard reporting. The analysis evaluates delivery performance, shipping modes, regional markets, customer segments, product profitability, and overall operational risk.

TOTAL ORDERS	TOTAL REVENUE	TOTAL PROFIT
180,519	$36.78M	$3.97M

LATE DELIVERY RATE	ON-TIME RATE	TOP MODE SHARE
54.83%	17.84%	59.69%

> **Primary Finding:** Late-delivery risk is the central operational issue facing DataCo. Shipping-mode analysis reveals **First Class** as the highest-risk mode with a **95.32% late delivery rate**, making fulfillment-process investigation an immediate business priority.

- **Volume Exposure:** Standard Class carries the largest order volume (59.69%), meaning process improvements here affect the vast majority of orders.
- **Performance Exposure:** First Class exhibits severe late-delivery rates, jeopardizing high-priority customer expectations.
- **Power BI Suite:** Delivered across three tailored dashboards: *Executive Overview*, *Shipping Performance*, and *Root Cause Drill-Down*.

2. Project Overview & Objectives

Business Problem: Management required deep visibility into delivery delay bottlenecks, shipping modes, regional market performance, and customer segment profitability to streamline supply chain operations.

- Identify delivery-risk patterns by shipping mode, market, product category, and timeframe.
- Quantify financial and operational impacts of recurring fulfillment failures.
- Utilize SQL aggregation for complex multi-variable business queries.
- Design an interactive Power BI dashboard suite catered to executive and operational stakeholders.

DATASET ATTRIBUTE	VALUE / SCOPE
Source	DataCo Global Supply Chain Dataset
Volume	180,519 Orders (2015–2018)
Markets	LATAM, Europe, Pacific Asia, USCA, Africa
Shipping Modes	Standard Class, Second Class, First Class, Same Day

3. Data Cleaning & Preparation

- **PVI Reductions:** Dropped unnecessary columns (*Product Description, Order Zipcode, Customer Email, Customer Password, Product Image, Customer Street*).
- **Missing Data:** Imputed missing `Customer Lname` values with *"Unknown"*.
- **Datetime Standardization:** Converted order and shipping date fields using mixed-format parsing to maintain precision.

```
import pandas as pd

df = pd.read_excel("DataCoSupplyChainDataset.PROJECT.xlsx")
df = df.drop(columns=["Product Description", "Order Zipcode", "Customer Email",
                      "Customer Password", "Product Image", "Customer Street"])

df["Customer Lname"] = df["Customer Lname"].fillna("Unknown")
df["order date (DateOrders)"] = pd.to_datetime(df["order date (DateOrders)"], format="mixed",
errors="coerce")
df["shipping date (DateOrders)"] = pd.to_datetime(df["shipping date (DateOrders)"],
format="mixed", errors="coerce")

df.to_excel("DataCoSupplyChain_Clean.xlsx", index=False)
```

Data-Quality Outcome: Zero duplicate rows and full validation before database ingestion.

4. Exploratory Data Analysis

DELIVERY STATUS	ORDERS	SHARE (%)
Late delivery	98,977	54.83%
Advance shipping	41,592	23.04%
Shipping on time	32,196	17.84%
Shipping canceled	7,754	4.30%

5. SQL Database Analysis

The clean dataset was ingested into MySQL for structured query analysis.

```
SELECT `Shipping Mode`,
       COUNT(*) AS Total_Orders,
       SUM(`Late_delivery_risk`) AS Late_Orders,
       ROUND(AVG(`Late_delivery_risk`) * 100, 1) AS Late_Pct
FROM datacosupplychain_clean_sql
GROUP BY `Shipping Mode`
ORDER BY Late_Pct DESC;
```

SHIPPING MODE	TOTAL ORDERS	LATE ORDERS	LATE RATE (%)
First Class	27,814	26,513	**95.32%**
Second Class	35,216	26,987	76.63%
Same Day	9,737	4,454	45.74%
Standard Class	107,752	41,023	38.07%

6. Power BI Dashboard Suite

DASHBOARD PAGE	AUDIENCE	MAIN PURPOSE
Executive Overview	C-Level / Management	High-level revenue, profit, order metrics & regional trends.
Shipping Performance	Logistics / Operations	Shipping delays, late order volume by mode and market.
Root Cause Drill-Down	Supply Chain Managers	Geographic breakdown, carrier handoff & market-mode matrix.

7. Recommendations & Action Plan

PRIORITY	RECOMMENDED ACTION	TIME HORIZON
Critical	Investigate First Class carrier performance, handoffs, and service level commitments.	Immediate
High	Trace fulfillment bottlenecks from picking to dispatch; establish SLA tracking.	30 Days
Medium	Audit negative-profit product categories (discounting & shipping cost rules).	60 Days
Medium	Implement priority retention monitoring for high-value enterprise customers.	Ongoing

8. Technical Summary

- **Python:** Pandas, NumPy data cleaning and exploratory data visualizations.
- **SQL:** Complex aggregation, windowed calculations, GROUP BY/HAVING operations.
- **Power BI:** DAX measures, relational modeling, dynamic slicers, drill-through layouts.